Knowledge management in family firms: a systematic review, integrated insights and future research opportunities

Emma Su and Joshua Daspit

Abstract

Purpose – *The literature related to knowledge management (KM) is robust with respect to insights regarding firms in general. However, less is known about the KM of family firms despite these firms being the most common form of business organization worldwide. Further, even though the number of studies examining family-firm KM has increased in recent years, the insights gained remain fragmented. Therefore, the purpose of this study is to help coalesce and advance the study of family-firm KM.*

Design/methodology/approach – *In pursuit of these goals, a systematic literature review was conducted. Using a 6-step, systematic literature review protocol, 74 articles focused on family-firm KM published in 23 journals were identified and reviewed.*

Findings – *This literature review contributes to the synthesis and advancement of family-firm KM scholarship in several ways. First, key factors and relationships are identified and integrated into a robust framework. Second, scholarly insights are synthesized, and a review of the primary antecedents, outcomes and moderating factors associated with family-firm KM processes is presented. Third, promising opportunities for future research are highlighted to advance family-firm KM scholarship.*

Originality/value – *With a focus on reducing the fragmentation in the literature, this review synthesizes insights related to the most commonly studied antecedents, outcomes and moderators associated with family-firm KM. Additionally, antecedents are organized and reviewed according to the nature of their influence on family-firm KM processes, highlighting the simultaneous opposite effects of some influences. Further, key outcomes are synthesized based on their family versus firm-centric orientation. Even further, insights and opportunities focused on advancing the theory, antecedents, outcomes, moderators and other issues related to family-firm KM are presented in an effort to support the continued progress of scholarship in this area.*

Keywords *Family firms, Knowledge management, Literature review, Future research directions*

Paper type *Literature review*

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1. Introduction

The "knowledge economy" is upon us. In recent years, scientists have increasingly noted the transition of advanced industrial nations from economies based on physical inputs to the increased reliance on knowledge-based assets (Latilla *et al.*, 2018; Powell and Snellman, 2004). Accordingly, knowledge is recognized as the most important resource in corporate operations (Singla *et al.*, 2020; Spender, 1996), and because of this, the ability of firms to strategically manage knowledge is paramount. Knowledge management (KM) – or the systematic management of knowledge implemented via processes [1] that are designed to facilitate the creation, storage, transfer and application of knowledge (Del Giudice *et al.*, 2010) – is vital for firm survival and success. While robust literature offers insight into factors associated with the KM of firms in general, less is known about the factors related to the KM of family firms specifically.

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Family firms are businesses owned and/or managed by a family or families with transgenerational succession intentions (Chua *et al.*, 1999). Family firms range in size and are the predominant form of business organization worldwide (Miller and Le Breton-Miller, 2005). Compared to nonfamily firms, family firms can have unique resources and capabilities (Habbershon *et al.*, 2003), governance configurations (Daspit *et al.*, 2018) and nonfinancial goals (Chrisman *et al.*, 2021; Gómez-Mejía *et al.*, 2007). These and other idiosyncratic characteristics, which result from the family's involvement in the firm, are known to influence KM processes in both positive and negative ways (Del Giudice *et al.*, 2010). The range of effects from these unique characteristics creates opportunities for investigating the emergence and influence of factors that are not easily (or ever) observed in nonfamily firms. Furthermore, while the number of studies on family-firm KM has increased in recent years, the insights gained remain fragmented. This fragmentation has resulted, in part, from a salient focus in the literature on understanding family-firm innovation [2], overshadowing the attention given to the underlying KM processes of family firms. Even further, measurement challenges – coupled with the conceptual complexity of family-specific characteristics – have also contributed to the fragmentation of the family-firm KM literature. If this fragmentation persists (or is exacerbated), it is likely to result in the inability to articulate differences among family firms, create theoretical inaccuracies, lead to empirical indeterminacies and limit the precision of practitioner recommendations (Chua *et al.*, 2012; Melin and Nordqvist, 2007).

Given that the knowledge of family-firm KM remains fragmented despite an increasing number of studies, we aim to help coalesce research in this area by reviewing the extant literature, synthesizing insights and offering directions for the advancement of scholarly work. To this end, we reviewed 74 articles on family-firm KM published in 23 journals. The findings of the review are integrated into a conceptual model consisting of antecedents, outcomes and moderators related to the primary KM processes. Further, to highlight the divergent nature of influences, we note both the positive and negative effects of the most commonly studied antecedents. In addition, outcomes are categorized according to their family-centric versus firm-centric orientation and the (moderating) boundary conditions on prominent relationships are identified.

This review offers several contributions to the literature. First, given the disjointed nature of scholarly work on family-firm KM, this review helps to coalesce academic insights by providing a synthesized understanding of the current state of knowledge. By integrating insights into a robust framework of antecedents, outcomes and moderating factors, we support the progress of knowledge that bridges gaps in this area of study. Second, given the idiosyncratic effects that result from the family's involvement in the firm, we elucidate both the positive and negative effects on KM, highlighting factors that have simultaneous opposite effects on KM. Further, we note that the effect of some factors varies depending on the KM process examined. Third, we contribute to the advancement of family-firm KM scholarship by synthesizing insights from the current literature and illuminating opportunities for future research.

2. Knowledge management processes

For the modern firm, the ability to appropriately manage knowledge is essential for survival and success. Given this importance, scholarly work in this area has become increasingly relevant as both researchers and practitioners seek to understand how best to systematically manage knowledge. This systematic management is often implemented via knowledge-related processes in the firm (Del Giudice *et al.*, 2010). While firms vary in how knowledge is managed, studies note that social and organizational dynamics are among the most salient factors that contribute to a firm's successful KM (Inkinen, 2016). Researchers generally agree that KM is a multi-component process involving knowledge-related activities that occur within and across the organization's boundaries for value-creation purposes (Davenport and Prusak, 1998;

Del Giudice and Maggioni, 2014; Durst and Edvardsson, 2012; Edvardsson, 2009). Given this consensus, we organize this review following the seminal work of Alavi and Leidner (2001), who developed a multi-process framework grounded in the sociology of knowledge.

Alavi and Leidner's (2001) framework acknowledges the social nature of KM by viewing firms as "social collectives" with "knowledge systems." The sociological underpinnings of this framework align with the idiosyncratic manifestations that result from the family's involvement in the firm. In other words, due to the family's involvement, family firms possess unique forms of social capital (Herrero, 2018), foster trust-based interpersonal exchanges (Long and Mathews, 2011), and in general, are unique given the integration of the family's relational dynamics in the firm (Habbershon *et al.*, 2003). This seminal framework by Alavi and Leidner (2001) complements the unique nature of the family firm; thus we follow this conceptualization and organize our review based on the four primary KM processes of knowledge creation, knowledge storage/retrieval, knowledge transfer and knowledge application.

2.1 Knowledge creation

Knowledge creation is the development of new knowledge that typically results in new ideas, inventions, practices and/or products, which includes replacing existing firm knowledge (Pentland, 1995; Phelps *et al.*, 2012). Knowledge creation often involves the interplay between tacit and explicit knowledge and is typically created through individuals' cognitive processes, as well as collaborative mechanisms (Nonaka, 1994). In other words, social interactions among organizational members can facilitate the creation of new knowledge as new insights are developed through communication; however, knowledge creation can also occur, for example, when current knowledge is combined, when tacit knowledge is converted to explicit knowledge (e.g. creating formalized practices) or when explicit knowledge is converted to tacit knowledge (e.g. individual learning from reading) (Nonaka, 1994).

2.2 Knowledge storage/retrieval

While firms create knowledge, they also have a "memory" and can "forget" (Argote *et al.*, 1990). The storage and retrieval of knowledge, or organizational memory (Walsh and Ungson, 1991), is integral to effective KM (Alavi and Leidner, 2001). Organizational memory is "[. . .] the means by which knowledge from the past, experience, and events influence present organizational activities" (Stein and Zwass, 1995: 85). Knowledge can be embedded for storage in various explicit and tacit forms, which include manuals, databases, culture and work processes (Moorman and Miner, 1998). In addition to storing knowledge, recalling the appropriate knowledge when needed – either from a knowledge repository or other storage system – allows stored knowledge to be retrieved for later use (Alavi and Leidner, 2001).

2.3 Knowledge transfer

Given its distributed nature, knowledge may require being transferred to a new location prior to application (Alavi and Leidner, 2001). Knowledge transfer can be affected by elements such as motivational disposition of the sending unit in terms of a willingness to share knowledge, motivational disposition of the receiving unit regarding a willingness to acquire knowledge from the source and the existence of transmission channels (Gupta and Govindarajan, 2000). Knowledge transfer can take place at various levels, which include between and/or among individuals, groups, organizations. Communication channels facilitate the transfer of knowledge through means that are formal (e.g. training sessions), informal (e.g. coffee-break conversations), personal (e.g. apprenticeships) and/or impersonal (e.g. knowledge repositories) in nature (Alavi and Leidner, 2001). The most

effective transfer mechanism also depends on the type of knowledge being transferred and the entities involved (Inkpen and Dinur, 1998).

2.4 Knowledge application

Knowledge itself does not necessarily directly lead to organizational outcomes; however, the effective application of knowledge is where commercial value is realized (Kogut and Zander, 1992). As noted by Grant (1996), effective knowledge application is typically governed by clear organizational directives (e.g. rules and procedures) with refined routines embedded in the firm that allow for application efficiency. Firm routines, which are developed based on coordination patterns and protocols, are important mechanisms for knowledge application because they allow individuals to integrate and use their knowledge without the need to communicate what they know to others (Alavi and Leidner, 2001). In short, without a systematic approach for applying knowledge within the firm, simply possessing stocks of knowledge yields marginal value, if any.

3. An overview of family-firm knowledge management

In the broader KM literature, most research focuses on KM processes in the context of large, professionalized, nonfamily firms (Gray, 2006). As a result, less is known, comparatively, about the factors that relate to these processes in family firms. Even though the number of family-firm KM studies continues to increase, the collective insights gained have remained limited despite the importance of managing knowledge for this type of organization: the most predominant form of enterprise worldwide (Miller and Le Breton-Miller, 2005).

While knowledge is valuable for all firms, family firms have characteristics that offer unique potential with respect to KM. For instance, family firms can generate deep levels of firm-specific tacit knowledge given the longer tenures of family firm leaders (Lee *et al.*, 2003; Sirmon and Hitt, 2003), and family firms are commonly considered long-term-oriented given their desire to pass the business on to future generations (Brigham *et al.*, 2014; Chua *et al.*, 1999), underscoring the importance of managing knowledge across generations. This transgenerational knowledge exchange includes the knowledge transferred from the incumbent to the successor, which is paramount for preparing the successor to lead the firm (Cabrera-Suárez *et al.*, 2018; Dou *et al.*, 2020; Woodfield and Husted, 2017). Family firms, being deeply rooted in traditions, also represent a particularly unique context for understanding organizational memory and how their past knowledge affects their present activities. For example, studies suggest that family firms are able to store prior knowledge and then retrieve this knowledge for the purpose of unique value creation, which is referred as "innovating through tradition" (De Massis *et al.*, 2016).

Additionally, family firms can have an advantage over their nonfamily counterparts with an enhanced relational flow among family members that facilitates knowledge transfer, especially the transfer of tacit knowledge (Cunningham *et al.*, 2016; Su and Carney, 2013). Further, the commitment to change among kin and their prior history together also foster the efficient exchange and application of knowledge (Brinkerink, 2018; Chirico and Salvato, 2008).

On the other hand, family firms also have characteristics that potentially limit KM processes. For instance, family firms tend to be more insular and less likely to engage in external knowledge exchange given the economies created among kin with the internal transfer of tacit knowledge (Patel and Fiet, 2011). Moreover, family firms are noted for their pursuit of socioemotional wealth (SEW; Gómez-Mejía *et al.*, 2007). The pursuit of these noneconomic goals can restrict the extent to which family firms share knowledge with nonfamily actors given the lack of trust and the risk of losing control (Brinkerink, 2018).

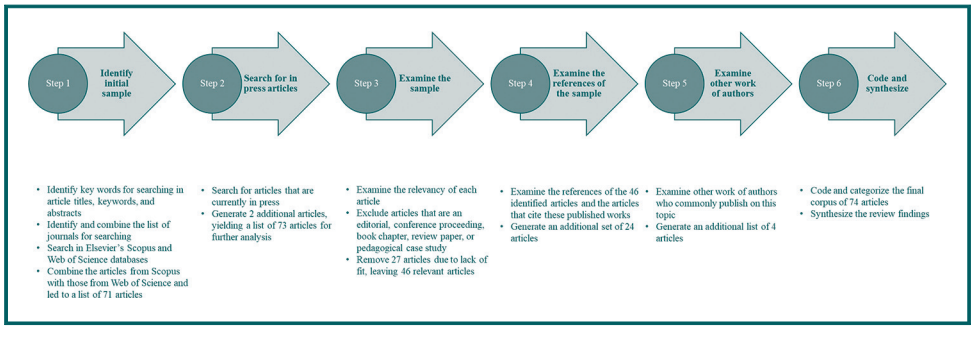
Nonetheless, while family firms, in general, exhibit unique behaviors when compared to nonfamily firms, not all family firms are the same. Specifically, studies have shown salient differences among family firms (see review by Daspit *et al.*, 2021). The generation in control, for instance, is a prominent factor that creates differences among family firms (Cruz and Nordqvist, 2012; Gersick *et al.*, 1997). Pittino *et al.* (2018), in their study of psychological ownership as an influence of entrepreneurial orientation (EO), show that the link between psychological ownership and EO is mediated through knowledge sharing. However, the effect of psychological ownership on knowledge sharing is not constant; the effect differs based on the family generation in control with the effect being weaker in later-generation family firms (Pittino *et al.*, 2018). The social ties among kin supported by bonding relationships and traditions are more prevalent in earlier-generation family firms, and these relational mechanisms facilitate knowledge-sharing activities. Furthermore, the effect is reduced by professionalization, which is more common in later-generation firms (Pittino *et al.*, 2018).

In addition to a range of influences observed in the literature, the fragmentation of family-firm KM scholarship is exacerbated by inconsistent conceptualizations of KM-specific processes and an overall lack of insights synthesized across disciplinary boundaries. The fragmentation in this area of study is similar to the fragmentation recently observed in the family-firm internationalization literature (see the review by Debellis *et al.*, 2021b). Like the scholarly work in that area, the study of family-firm KM suffers from a lack of consistency in its theoretical and empirical underpinnings. Therefore, we proceed with the objective to help coalesce insights through a systematic review of the literature.

4. Scope of the review

To conduct a comprehensive review of the literature, we followed a systematic approach used by Stephan (2018) that is based on the work of Tranfield *et al.* (2003). Our systematic approach includes a six-step protocol (Figure 1 provides an overview). First, following the review of Debellis *et al.* (2021b), we conducted a broad search (not restricted by date) of article titles, keywords and abstracts using a comprehensive set of search terms [3]. To ensure a robust search, we conducted the search using the following two commonly used databases: Scopus and Web of Science (WoS) [4] (Hernández-Linares and López-Fernández, 2018). The searches were conducted to find content published in major management and organization-focused journals using the journal-selection criteria used by Debellis *et al.* (2021b). Accordingly, we combined the *Financial Times* top-50 journals with journals ranked as "3" or above in the 2018 ABS Academic Journal Guide that were categorized as part of the following subject areas: business history and economic history; entrepreneurship and small business management; general management, ethics, gender and social responsibility; international business and area studies; innovation; marketing;

Figure 1 Article selection process



organization studies; and strategy. In addition, the *Journal of Knowledge Management* and *Journal of Family Business Strategy* were included given the relevance of these journals to the review topic [5]. The search, which was conducted within a total of 110 journals, identified 36 articles from the Scopus database and 64 articles from the WoS database [6]. The identified articles were then combined and reviewed for duplicates, resulting in 71 articles.

In the second step, "in press" articles were identified. To this end, articles that were available ahead of publication (i.e. articles made available after journal acceptance but prior to publication in a dedicated journal issue) were searched using the same search-related criteria noted in the previous step. This search was conducted in all 110 journals by visiting each website directly. This step led to identifying 2 additional articles, bringing the total to 73 articles.

In the third step, we reviewed each article in more detail following the work of Andreini *et al.* (2020). To be included in the review, each article was required to examine at least one KM process [7]. Further, the article was required to examine a KM process in the context of family firms or in the case of comparison studies, compare such process between family and nonfamily (or potentially other) firms. Editorials, conference proceedings, book chapters, review articles and pedagogical studies were excluded. To provide a thorough review, conceptual articles were included. In all, 27 articles were removed due to lack of fit, leaving 46 relevant articles.

In the next two steps, we sought to ensure the robustness of the search. Specifically, in the fourth step, following Schnatterly *et al.* (2018), we examined the references of the 46 identified articles and reviewed studies that directly cite these articles. For inclusion, the title, abstract, keywords, and, when necessary, the full study were examined to ensure compliance with the aforementioned inclusion criteria. This resulted in 24 new articles being identified.

In the fifth step, the authors who commonly publish on this topic were identified and their other publications were examined. Following the protocol of Aguinis and Glavas (2012), we identified authors with at least two publications who were included in our article pool. Then, the publication record of each identified author was examined with the intention of identifying additional, relevant work. This process led to the addition of 4 articles that met the inclusion criteria. Thus, the final corpus includes 74 articles that are published in 23 journals [8].

Finally, in the sixth step, each article was coded and categorized. In addition to capturing basic identifying information for each article, we coded the following components when relevant: the article's research question(s); whether a conceptual or empirical contribution; the primary KM process(es) examined; antecedent(s), outcome(s) and moderator(s) related to the KM process(es); theoretical framework; details on the sample and method(s) used for empirical studies; main finding(s); and other data. Table 1 provides a summary of the corpus of articles.

5. Organizing framework and review findings

Given the developing nature of work in this area, we report on the key constructs and relationships identified from our review of family-firm KM scholarship. These key findings are integrated using an organizing framework of antecedents, outcomes and moderators related to the primary KM processes (for similar frameworks, see: Combs *et al.*, 2011; Kwon *et al.*, 2020; Wan *et al.*, 2011). Figure 2 displays the organizing framework and key constructs. The findings of the review are presented below with a focus on the primary antecedents, outcomes and moderating factors. Finally, integrated insights and future research opportunities are offered.

Table 1 Summary of articles included in the review of family-firm KM

Author(s) and year	*Empirical/ conceptual*	*Theory/perspective*	*Research area(s)*	*Primary KM process(es)*
Aiello *et al.* (2020)	Empirical	SEW; family social capital	2	Knowledge transfer
Ayyagari *et al.* (2011)	Empirical	Not specified	1	Knowledge transfer
Baday Yıldız *et al.* (2021)	Empirical	Behavioral theory of the firm; dynamic capabilities	2	Knowledge application
Bamel and Bamel (2018)	Empirical	RBV; dynamic capabilities	2	Knowledge application
Barros *et al.* (2016)	Conceptual	Dynamic capabilities; SEW	2	Knowledge application
Basco (2015)	Conceptual	Absorptive capacity	2	Knowledge application
Biscotti *et al.* (2018)	Empirical	KBV; identity	1,2	Knowledge creation
Boh *et al.* (2020)	Empirical	Organizational learning; SEW	2	Knowledge application
Bouncken *et al.* (2020)	Empirical	KBV	1,3	Knowledge creation
Boyd and Hollensen (2012)	Empirical	Absorptive capacity	2	Knowledge application
Brinkerink and Rondi (2021)	Empirical	Institutional theory	1	Knowledge storage/retrieval
Brinkerink (2018)	Empirical	Absorptive capacity; organizational learning	1	Knowledge application, creation
Cabrera-Suárez *et al.* (2011)	Conceptual	RBV; stakeholder theory	2,3	Various KM processes
Cabrera-Suárez *et al.* (2011)	Conceptual	KBV; RBV	1,2	Knowledge transfer
Cabrera-Suárez *et al.* (2018)	Conceptual	RBV; KBV	1	Knowledge creation
Casprini *et al.* (2017)	Empirical	KBV	2	Knowledge transfer, application
Cesinger *et al.* (2016)	Empirical	Uppsala model; SEW	1,2	Knowledge transfer
Chaudhary and Batra (2018)	Empirical	Absorptive capacity; KBV	2	Knowledge application
Chirico and Nordqvist (2010)	Empirical	Dynamic capabilities	2	Various KM processes
Chirico and Salvato (2008)	Conceptual	KBV	1,2	Knowledge application
Chirico and Salvato (2016)	Empirical	KBV	2,3	Knowledge creation
Chirico (2008)	Empirical	KBV	1	Knowledge transfer, application
Chirico *et al.* (2011)	Empirical	Resource orchestration	2	Knowledge transfer
Cunningham *et al.* (2016)	Empirical	Path-goal theory	1	Knowledge transfer
Cunningham *et al.* (2017)	Empirical	KBV	1,3	Knowledge transfer
Daspit *et al.* (2019)	Conceptual	Dynamic capabilities; absorptive capacity; familiness	1,2,3	Knowledge application
De Massis *et al.* (2018)	Empirical	RBV	2	Knowledge application
Debellis *et al.* (2021a)	Conceptual	Ability and willingness perspective	1,2	Knowledge transfer, application
Debicki *et al.* (2020)	Empirical	Social capital; SEW	2,3	Knowledge application
Dou *et al.* (2020)	Empirical	Social learning theory	2	Knowledge transfer
Erdogan *et al.* (2020)	Empirical	Family imprinting perspective	2	Knowledge storage/retrieval
Fang *et al.* (2018)	Empirical	KBV	2	Knowledge application
Fernández Moya (2010)	Empirical	Not specified	1,2	Knowledge transfer
Giovannoni *et al.* (2011)	Empirical	KBV	1	Knowledge transfer
Glyptis *et al.* (2021)	Empirical	Dynamic capabilities; RBV	1	Knowledge application
Hatak and Roessl (2015)	Empirical	KBV	1	Knowledge transfer
Hernández-Perlines *et al.* (2017)	Empirical	Absorptive capacity	2	Knowledge application
Herrero and Hughes (2019)	empirical	Social capital; family social capital	2,3	Knowledge transfer
Howorth *et al.* (2004)	Conceptual	Agency theory	1,2	Knowledge transfer
Jaskiewicz *et al.* (2013)	Conceptual	Social exchange theory	1,2	Knowledge transfer
Jones and Craven (2001)	Empirical	Absorptive capacity	1	Knowledge application
Kammerlander *et al.* (2015)	Empirical	KBV; path dependence	2	Knowledge transfer, storage/ retrieval

(*continued*)

Table 1

Author(s) and year	Empirical/ conceptual	Theory/perspective	Research area(s)	Primary KM process(es)
Kotlar *et al.* (2020)	Conceptual	Absorptive capacity	1,3	Knowledge transfer, application
Kotlar *et al.* (2013)	Empirical	Behavioral agency model	1,3	Knowledge transfer, application
LeCounte (2020)	Conceptual	KBV	2	Knowledge transfer
Lee *et al.* (2003)	Conceptual	Game theory	2	Knowledge transfer
Madison *et al.* (2021)	Empirical	Transactive memory systems theory	2	Knowledge storage/retrieval
Magistretti *et al.* (2020)	Empirical	KBV	2	Knowledge storage/retrieval
Manzaneque *et al.* (2020)	Empirical	Behavioral theory of the firm	1,2,3	Knowledge transfer, application
Mazzola *et al.* (2008)	Empirical	Grounded theory	1	Knowledge creation
Ng and Keasey (2010)	Empirical	Absorptive capacity	2	Knowledge application
Patel and Fiet (2011)	Conceptual	Family business dynamics	1,2	Knowledge creation, application
Perez and Puig (2004)	Conceptual	New institutionalism and sociological perspectives	2	Knowledge transfer
Pittino *et al.* (2018)	Empirical	KBV; stewardship theory	2,3	Knowledge transfer
Randolph *et al.* (2021)	Empirical	Social capital; causation and effectuation logics	2	Knowledge transfer
Randolph *et al.* (2017)	Conceptual	KBV; SEW	2	Knowledge transfer
Royer *et al.* (2008)	Empirical	TCE	2	Knowledge transfer
Salvato and Melin (2008)	Empirical	Social capital	2	Knowledge application
Sánchez-Sellero *et al.* (2014)	Empirical	Absorptive capacity	1	Knowledge application
Santoro *et al.* (2021)	Empirical	Resiliency and psychological capital perspectives	2,3	Knowledge transfer, creation and application
Savino *et al.* (2017)	Empirical	Not specified	2	Knowledge creation, storage/ retrieval
Sciascia and Mazzola (2008)	Empirical	Agency theory; stewardship theory; KBV	2	Knowledge transfer
Sciascia *et al.* (2013)	Empirical	Upper echelons theory	2	Knowledge transfer
Soluk *et al.* (2021)	Empirical	Dynamic capabilities	1,2,3	Knowledge application
Su and Carney (2013)	Empirical	KBV	1	Knowledge transfer, creation
Su and Dou (2013)	Empirical	KBV	2	Knowledge transfer
Tsang (2020)	Conceptual	Organizational learning	2	Knowledge application
Werner *et al.* (2018)	Empirical	KBV; SEW	1,2	Knowledge storage/retrieval
Woodfield and Husted (2017)	Empirical	KBV	1	Knowledge transfer
Young and Tsai (2008)	Empirical	Social capital	2	Knowledge transfer
Zahra (2012)	Empirical	Behavioral theory of the firm	1,2,3	Knowledge application, transfer
Zahra *et al.* (2004)	Empirical	RBV	1,2	Knowledge creation, transfer
Zahra *et al.* (2007)	Empirical	KBV	2,3	Knowledge transfer
Zybura *et al.* (2021)	Empirical	RBV; KBV	2	Knowledge transfer

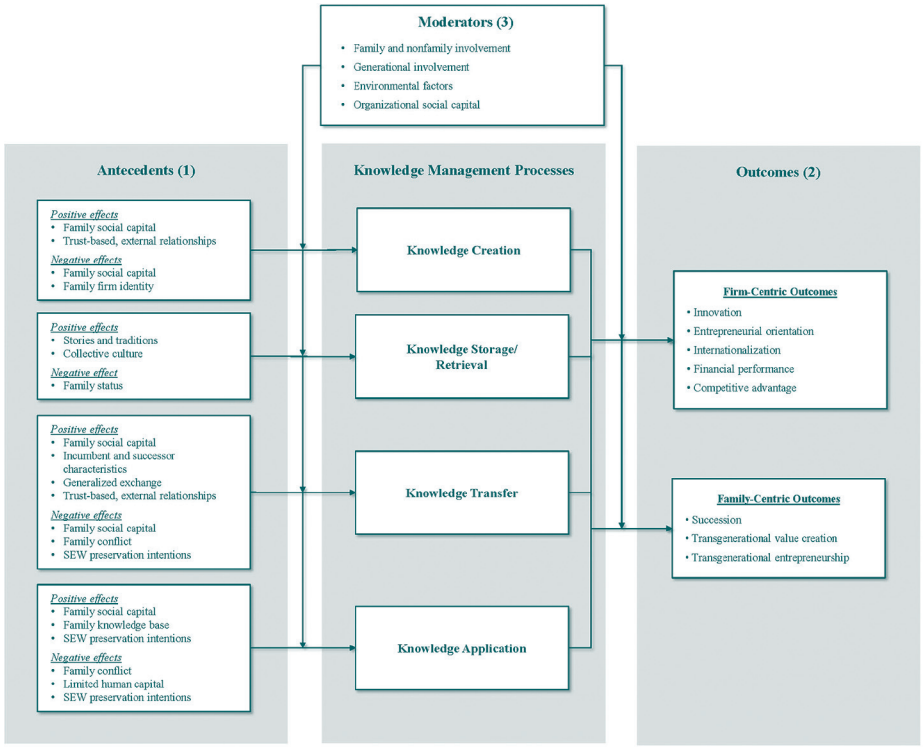
Notes: For the research area(s), 1 = Antecedents, 2 = Outcomes, 3 = Moderators, as delineated in Figure 2; RBV: Resource-based view; KBV: Knowledge-based view; SEW: Socioemotional wealth

5.1 Antecedents of knowledge management processes in family firms

Our review of family-firm KM finds that, relatively, a notable amount of research has been conducted on the antecedents of KM processes in family firms. Given the unique nature of the family's involvement in the firm, our review finds that antecedents – especially those that are family-specific – can be categorized based on the nature of their influence: positive and/or negative effects on KM. To this end, for each KM process, we categorize antecedents based on the nature of their influence, focusing on the most commonly investigated antecedents.

5.1.1 Knowledge creation. 5.1.1.1 Positive effects. When the family is involved in the firm, a unique bundle of resources and capabilities manifest that are distinct to family firms (Habbershon and Williams, 1999; Habbershon *et al.*, 2003). Although this "familiness" includes a range of unique, family-influenced features, one of the resources most commonly

Figure 2 Conceptual map of family-firm KM research



Notes: The numbers in parentheses correspond to the primary research area of each article noted in Table 1. The conceptual map corresponds to the review offered, which focuses on the key antecedents, outcomes and moderators observed in the literature

studied for its effects on knowledge creation relates to a form of social capital known as family social capital. *Family social capital* [9] consists of the unique resources embedded in and accessible through the network of family-based relationships (Arregle *et al.*, 2007). Family social capital is a particular type of bonding social capital, steeped in deep levels of trust, which exists among family members in the family firm (Herrero, 2018).

Family social capital affects the family firm's knowledge creation. This form of social capital among kin has the potential to deliver a knowledge-creation advantage for family firms (compared to nonfamily firms) given the ability of family firms to efficiently synthesize existing knowledge via their unique relational bonds (Patel and Fiet, 2011). The strong bonds associated with family social capital enable a more creative (re)combination of new and existing knowledge than what is likely to otherwise exist. Brinkerink (2018), for example, finds that family firms are more capable of being "closed innovators" due, in part, to these unique relational bonds. In fact, this relational closeness among kin, which reinforces "traditional" knowledge, helps enhance the family firm's ability to combine and create new knowledge across generations (Brinkerink, 2018; De Massis *et al.*, 2016).

While acknowledging the value of family bonds, Patel and Fiet (2011) underscore the family firm's long-term orientation and the shared common language among family members as these factors – which support the development of family social capital – also promote stewardship behavior and limit turnover, allowing the family firm to develop more complex cognitive systems and, ultimately, create more new knowledge. The shared collective knowledge among kin is more likely to be combined to create new, internal knowledge

when family social capital exists, yet organizational mechanisms can also be used to facilitate knowledge creation. One example is the use of family meetings during which existing and new knowledge can be combined (potentially across generations) to generate new insights (Zahra *et al.*, 2004).

Family social capital enhances knowledge creation within the family firm owing to the noted benefits; however, knowledge creation also occurs beyond the firm's boundary. Specifically, studies highlight the effect of external relationships on the family firm's knowledge creation process and find that family firms are cautious when engaging with external organizations. When knowledge creation opportunities that involve an external organization are identified, family firms are more likely to engage when *trust-based, external relationships* exist with such organizations (Arregle *et al.*, 2007; Miller and Le Breton-Miller, 2005). Based on insights drawn from the dynamic relational view, Bouncken *et al.* (2020) show that when engaging in knowledge-creation activities with external firms, family firms rely on trust (rather than contracting) with alliance partners, and further, heavily specified contracts are damaging to knowledge creation (attributed to the lack of trust); whereas nonfamily firms engage with alliance partners when either trust or a contract exists.

5.1.1.2 Negative effects. While some studies find that family social capital enhances knowledge creation in the family firm, others note that family social capital has negative effects on knowledge creation. For instance, Su and Carney (2013) suggest that when *family social capital* is salient, nonfamily employees are more likely to be perceived as less trustworthy. Nonfamily employees – who are a potentially rich source of diverse knowledge for the family firm given that family members tend to share similar knowledge and have overlapping networks – may not contribute the full scope of their knowledge given the lack of trust from (or with) the family firm. Ultimately, this results in a potentially impactful hindrance on the family firm's knowledge creation ability.

Furthermore, despite the benefits of the unique factors that result from the family's involvement in the firm, the extent to which family members identify with the firm is noted to impact knowledge creation, creating differences among family firms. Specifically, drawing on insights from social identity theory, Biscotti *et al.* (2018) suggest that the *family firm identity* held by a member of the controlling family can create conflict between family-centered priorities and the needs of the family business. This conflict results in family owners underestimating the firm's learning-related needs and attributing less importance to activities such as employee training and development programs. Such actions negatively impact employee development and limit knowledge creation in the family firm. In all, studies collectively show that the involvement of the family in the firm both enhances and limits knowledge creation through various influences.

5.1.2 Knowledge storage/retrieval. 5.1.2.1 Positive effects. Organizations, as noted, store knowledge for later retrieval and use. The process of storing and retrieving knowledge allows knowledge from past experiences to be applied to present endeavors (Stein and Zwass, 1995). The storage and retrieval of knowledge, or an organization's memory, manifests via language, interactions and stories (Walsh and Ungson, 1991). Broadly, studies noting positive effects on family-firm knowledge storage/retrieval focus on the influence of stories and traditions, as well as the influence of the firm's collective culture.

The *stories and traditions* of the family firm are distinctive mechanisms for storing and retrieving knowledge that potentially create transgenerational value (Erdogan *et al.*, 2020). As noted by Kammerlander *et al.* (2015), the core of family-centered stories remains quite stable over time, further underscoring that shared stories are a worthwhile means of long-term knowledge storage for family firms. However, long-established family firms with a rich endowment of stories and traditions create a paradox given the simultaneous need for continuity and change. Interestingly, Erdogan *et al.* (2020) address this tension and introduce the construct of "temporal symbiosis," which allows the family firm to concurrently preserve their rich traditions while recalling and leveraging stored knowledge to remain

current. In other words, achieving this balance allows family firms to retain their history without becoming irrelevant.

The family firm's enhanced organizational memory is also a result of a *collective culture*. A collective and "people-centric" culture, for which many family firms are known, includes a long-term commitment to employees and enhanced trust (Werner *et al.*, 2018), which often results in less turnover – especially during periods of market decline – compared to nonfamily firms (Lee, 2006). Less turnover, in the context of a collective culture underpinned by long-term commitment and trust, supports knowledge storage given that long-serving employees are a knowledge repository (Grant, 1996; Posch and Wiedenegger, 2014). Offering further positive support for a collective orientation, Brinkerink and Rondi (2021) demonstrate that the loyalty and commitment exhibited when the family is involved in the firm decreases turnover and acts as a safeguard toward the storage of knowledge assets, limiting the leak of sensitive knowledge.

5.1.2.2 Negative effects. Studies also find that family-specific influences, like *family status*, can hinder knowledge storage and retrieval in family firms. For instance, Madison *et al.* (2021) examine the transactive memory system of the family firm: a firm-level system for storing and retrieving knowledge that resides among individuals (Ren and Argote, 2011; Wegner *et al.*, 1985). Using triadic data from family employees, nonfamily employees, and leaders in family firms, the findings show that the communication between family and nonfamily employees hinders, rather than enhances, the shared perceptions of where knowledge is stored within the transactive memory system. With further analyses, Madison *et al.* (2021) show that the unexpected finding results from communication having differing effects based on family status; specifically, for nonfamily employees, communication enhances firm-level innovation but has a negative effect on shared perceptions of knowledge storage. Overall, studies such as this underscore the importance of the type, quantity and quality of communication between family and nonfamily employees given the effects on how knowledge is stored and retrieved when (family status) differences exist.

5.1.3 Knowledge transfer. 5.1.3.1 Positive effects. Knowledge transfer is the most studied KM process in family firms, and of the knowledge-transfer antecedents examined, *family social capital* is one of the most common themes. As a resource among kin in the firm, this unique form of social capital – and the deep, trust-based relational bonds with which it is associated – can positively affect knowledge transfer in the family firm. A contingent of studies that examines knowledge transfer focus, specifically, on transgenerational knowledge transfer (Cabrera-Suárez *et al.*, 2001; Giovannoni *et al.*, 2011; Hatak and Roessl, 2015; LeCounte, 2020). Transgenerational knowledge transfer is often examined in the context of succession, which is perhaps unsurprising given that succession is one of the most commonly studied topics in family business research (De Massis *et al.*, 2008).

Relationships are vehicles of knowledge exchange; when relational bonds are strong in the succession process, knowledge transfer is enhanced (Daspit *et al.*, 2016). The valuable effect of familial ties is highlighted, for example, in a case study of a prominent historical publishing house in Spain. In this study, Fernández Moya (2010) shows that the intimacy of the father-son relationship enables transgenerational knowledge transfer. Similar findings related to transgenerational knowledge transfer hold across cultural and geographic contexts. Studies of family firms in China, for instance, demonstrate that interactions among family members, along with the next generation's early firm involvement, enrich transgenerational knowledge transfer in a family firm (Su and Carney, 2013). Indeed, consensus exists that family social capital can facilitate the transfer of knowledge (Cunningham *et al.*, 2016, 2017).

Further, the transgenerational knowledge transfer between the successor and incumbent is a focus of studies with the primary influences governing this knowledge transfer relating to *incumbent and successor characteristics*. While numerous characteristics of both the incumbent and successor have been examined, among the studies reviewed, a common

focus relates to the importance of the successor's competence. In particular, the perceived relational competence of the successor, in terms of the successor's ability to initiate and maintain relationships, is positively related to the accuracy and amount of knowledge transferred from the incumbent to the successor (Hatak and Roessl, 2015). The successor's relational competence is used as a signal to indicate whether they are likely to behave in a trustworthy manner, which mitigates the incumbent's uncertainty and enhances the willingness of the incumbent to share better-quality (and more) knowledge with the successor (Hatak and Roessl, 2015).

Related to family social capital and trust, attention is also given to familial relationships from a social exchange perspective. In their review of family-firm succession, Daspit *et al.* (2016) note the benefits (and limitations) of generalized versus restricted social exchanges among the stakeholders involved in succession and, specifically, highlight that *generalized exchange* between and among stakeholders (i.e. collectively focused exchange with no immediate expectation of return) benefits knowledge transfer through the phases of succession. Further, Jaskiewicz *et al.* (2013) suggest that generalized exchange is enhanced by reciprocal nepotism, which they posit is distinct from entitlement nepotism. Reciprocal nepotism – or nepotism developed from family-member interdependence, prior interactions among kin and family-benefitting norms – is theorized to enhance the transfer of tacit knowledge through the generalized exchange relationship between the hiring family member and the family member benefitting from the nepotistic treatment (Jaskiewicz *et al.*, 2013).

While relationships in the family firm are important to internal knowledge transfer, the transfer of knowledge also occurs across the firm's boundary. When such transfer occurs, having trust with the exchange partner is crucial for most family firms. For instance, in a study of management buy-outs and buy-ins, Howorth *et al.* (2004) show that *trust-based, external relationships* reduce information asymmetry and facilitate the transfer of tacit knowledge between the parties involved. Similar views are supported by Cesinger *et al.* (2016), who note that trust and commitment enhance knowledge transfer with external parties, owing to the family firm perceiving less risk of resource exposure due to the trust with the exchange partner. Overall, cross-boundary knowledge transfer is enhanced when the family firm trusts the exchange partner.

5.1.3.2 Negative effects. While family firms have characteristics that enhance knowledge transfer, not all characteristics yield positive effects. For instance, although nonfamily employees are an important source of knowledge for family firms (Randolph *et al.*, 2021; Tabor *et al.*, 2018), knowledge transfer is less likely to occur between family and nonfamily employees when *family social capital* is salient. When a large proportion of family members are involved in the business, and when those family members are linked by strong familial ties, knowledge transfer with nonfamily members is less likely to occur (tacit knowledge in particular) in such conditions (Su and Carney, 2013). Strong family ties represent intensive interactions within the family and reduce the exposure and the motivation to engage with outsiders like nonfamily employees (Daspit *et al.*, 2019). In other words, familial ties can be "too much of a good thing" (Herrero and Hughes, 2019) by creating insularity among kin that results in less motivation to share knowledge with nonfamily members (Su and Carney, 2013).

In some family firms, conflict exists among kin. A strong family influence in the firm may support family social capital; however, the same family influence may also cause *family conflict*, which consists of, but is not limited to, interpersonal tension, affective differences and sibling rivalry. Although Kellermanns and Eddleston (2004) note that conflict can be beneficial for the family firm, they also note that some types are destructive. For instance, relational conflict negatively affects family-firm performance due, in part, to the hindrance on interpersonal exchange and mutual understanding (Eddleston and Kellermanns, 2007).

Furthermore, while trust-based relationships help to facilitate knowledge transfer between the family firm and external actors (Cesinger *et al.*, 2016; Howorth *et al.*, 2004), the extent to

which a family firm engages in such external exchanges is often influenced by the *SEW preservation intentions* of the firm's controlling family. The intentions of the controlling family to preserve SEW (i.e. nonfinancial aspects or affective endowments of family owners; Gómez-Mejía *et al.*, 2007) are related to the family's emotional attachment to the firm, power concentration, identification with the firm and related factors (Berrone *et al.*, 2012). The intentions to preserve such elements are noted to restrict knowledge transfer between the family firm and external (nonfamily) parties (Debellis *et al.*, 2021a; Kotlar *et al.*, 2020; Manzaneque *et al.*, 2020). For example, Kotlar *et al.* (2020) argue that when family owners have a strong emotional attachment to their business, they are more likely to perceive knowledge transfer with external, nonfamily parties as a threat to maintaining their SEW, thus suggesting that emotional attachment limits external knowledge exchange. Likewise, Manzaneque *et al.* (2020) show that the family's concern over discretionary power loss to external actors (with whom they have limited trust) reduces their tendency to engage in knowledge transfer with external parties. In all, while varied effects are observed to positively and negatively influence the family firm's knowledge transfer, many of the influences have a relational and/or trust-based orientation.

5.1.4 Knowledge application. 5.1.4.1 Positive effects. From the knowledge-based perspective of the firm, the source of a firm's competitive advantage resides in its knowledge application process, or rather, its ability to apply (exploit) knowledge for commercial purposes (Alavi and Leidner, 2001; Grant, 1996). To this point, our review has shown that family social capital affects knowledge creation and knowledge transfer. Similarly, studies note that family social capital enhances knowledge application.

During the knowledge application process, new knowledge is applied to firm operations: a process that is complex and one that requires patience along with a detailed understanding of the firm and its processes (Gold *et al.*, 2001; Kogut and Zander, 1992). Family firms with more established *family social capital* are more likely to possess valuable tacit knowledge about the firm's processes and have lower governance costs, allowing these family firms to excel at applying new knowledge (Glyptis *et al.*, 2021). Governance costs are lessened as a result of the "socially intense" and trust-based relationships among family members; in turn, the refined coordination among kin allows for enhanced integration and application of their individual, specialized knowledge (Chirico and Salvato, 2008). Likewise, given that family social capital yields collective goals and actions, kin in the firm are more likely to share the desire to apply knowledge for the benefit of the collective (Daspit *et al.*, 2019; Pearson *et al.*, 2008).

In addition to the distinctive relationships that constitute family social capital, the shared experiences, frequent exchanges and length of time affiliated with the family firm contribute to a specialized *family knowledge base* embedded in the tacit knowledge of kin. Family firms (compared to nonfamily firms) are shown to excel in their ability to process and exploit internally developed knowledge (Brinkerink, 2018). This benefit is attributed to several factors that include the synergistic perspectives among family members given that they have differing, yet complementary, educational and professional backgrounds. A knowledge base composed of complementary resources among kin enhances how knowledge is orchestrated and, thereby, enhances the application of knowledge in the family firm (Chirico, 2008; Soluk *et al.*, 2021).

Further, a family firm's ability to apply knowledge is associated with their *SEW preservation intentions*. For example, Kotlar *et al.* (2020) suggest that the family owner's emotional attachment can be a strong incentive to apply knowledge. When a strong emotional attachment to the firm exists, family owners are more motivated to ensure knowledge is properly applied because their name and reputation are at stake (Kotlar *et al.*, 2020). Therefore, controlling family members with the desire to preserve their SEW enhance the family firm's knowledge application due to their emotional investment in the firm.

5.1.4.2 Negative effects. Even though knowledge application can be positively influenced by the aforementioned factors, the process can also be hindered. Among the key factors that restrict knowledge application is *family conflict*. Family firms can be fertile environments for conflict, and depending on the family's culture, familial conflict may be common (Kellermanns and Eddleston, 2004). Sibling rivalry and other forms of conflict generated from interpersonal, affective incompatibilities prevent family members from integrating knowledge, weakening the effectiveness and efficiency of knowledge application in the firm (Chirico and Salvato, 2008).

Additionally, family firms often have *limited human capital* given that they select from a smaller pool of talent given the types of employees generally attracted to work in a family firm (Fang *et al.*, 2021). The limited talent pool potentially restricts the quality of nonfamily members hired, and if nonfamily members are hired who do not enhance the human capital of the firm, the ability for the family firm to successfully engage in knowledge application may be jeopardized (Sánchez-Sellero *et al.*, 2014; Zahra, 2012). On the other hand, if nonfamily members with diverse knowledge resources are involved in the family firm, the goals of the nonfamily members may differ from the dominant coalition of family and/or nonfamily employees may not be well-socialized within the family firm. Moreover, the strong social ties among family members can create a boundary between family and nonfamily members, further exacerbating the division between family and nonfamily employees, and such a faction between family and nonfamily employees limits the application of knowledge (Daspit *et al.*, 2019).

Even further, while the family owner's emotional attachment is noted to increase knowledge application in the family firm (Kotlar *et al.*, 2020), studies also suggest that other *SEW preservation intentions* negatively affect knowledge application. In particular, family firms may encounter difficulties extracting value from external, exploratory knowledge given the family's desire to maintain control of the firm (Brinkerink, 2018). Offering a similar conclusion, Kotlar *et al.* (2020) posit that the family owner's intention to maintain concentrated power limits the application of knowledge due to fear of losing control of the firm. Overall, in family firms, numerous influences are observed to positively and/or negatively affect knowledge application.

5.2 Outcomes of knowledge management processes in family firms

Key family-firm KM outcomes that have been studied are noted. Following Holt *et al.* (2017), we categorize outcomes as firm- and family-centric. Firm-centric outcomes "represent meaningful outcomes for the firm that, while they may be affected by the family system, would also play a role in firms with no family involvement" while family-centric outcomes "represent meaningful outcomes for the family that may be affected by the business system" (Holt *et al.*, 2017: 186).

5.2.1 Firm-centric outcomes. 5.2.1.1 Innovation. Given that KM is essential for firm innovation (Nonaka and Takeuchi, 1995; Spender, 1996), it is not unexpected that the most commonly studied outcome is *innovation*. For family firms, innovation is critical to long-term sustainability (De Massis *et al.*, 2013; Rondi *et al.*, 2019) and is necessary for realizing their transgenerational succession intentions, which is a defining component of the family firm (Chua *et al.*, 1999).

The family's capability to efficiently combine and apply knowledge creates a unique advantage over nonfamily firms in identifying and exploiting opportunities related to innovation (Patel and Fiet, 2011; Zahra, 2012). For example, Patel and Fiet (2011) argue that noneconomic goals, such as a long-term sustainability intention, help family members to view the growth of their firm as a collective family responsibility, and with a more collective orientation, internal knowledge transfer is enhanced, which leads to the identification of more opportunities and enhances firm innovation. Similarly, the presence of familiness and

family social capital enhances the transformation and application of knowledge, thus leading to greater innovation (Aiello *et al.*, 2020; Baday Yıldız *et al.*, 2021; Daspit *et al.*, 2019). Even though some family firms acquire less knowledge from external parties (Kotlar *et al.*, 2013, 2020), they may have a refined capacity to translate external knowledge into tangible benefits and improve innovation (Aiello *et al.*, 2020). Moreover, Baday Yıldız *et al.* (2021) show that the family firm's innovation performance is positively driven by specialized tacit knowledge within the firm and (trusted) collaborations with external stakeholders. Together these findings suggest that the closeness among kin can enhance internal knowledge development (and even external collaborations when engaged with a trustworthy stakeholder), which enhances innovation outcomes of the family firm (Baday Yıldız *et al.*, 2021).

As family firms are often deeply rooted in their traditions, knowledge from the past may be leveraged to influence their present activities. Family firms are in a privileged position for leveraging their traditions to innovate. "Innovating through tradition" provides interesting insights on how knowledge from the past influences innovation in family firms (De Massis *et al.*, 2016). Even though firms that follow their traditions are typically thought to suffer from inertia and develop core rigidities, recent work has shown that family firms can leverage knowledge stored in their traditions to support innovation, highlighting the potential advantages of past knowledge for the family firm's innovation pursuits (De Massis *et al.*, 2016). In fact, Magistretti *et al.* (2020) show that family firms are better able to store and manage "old" knowledge, which puts them in a better position to establish links with tradition. "Innovating through tradition," though, creates challenges, yet family firms with a "temporal symbiosis" capability – or a capability to simultaneously adopt retrospective and prospective approaches that guide resource use – are more likely to preserve traditions while innovating (Erdogan *et al.*, 2020).

In addition to using the past to innovate, family firms face unique barriers when executing an open-innovation strategy. The family firm's desire for cohesiveness and control can lead to a decreased emphasis on acquiring external knowledge and on creating new external relationships, which restricts the quality and quantity of innovation-related outcomes (Boh *et al.*, 2020). Based on an exploratory case study of an Italian family firm, Casprini *et al.* (2017) identify the following two idiosyncratic capabilities that family firms use to overcome such barriers: imprinting capability and fraternization capability. The imprinting capability supports open-innovation strategies by improving knowledge transfer from the family to the business system and allowing the family values to be better understood; whereas the fraternization capability supports innovation strategies by building trust in social interactions and developing stable and long-lasting relationships with employees and outside partners (Casprini *et al.*, 2017).

Finally, research also shows that innovation outcomes are related to the nature of knowledge shared across generations. For example, applying a multiple-case-study research design, Kammerlander *et al.* (2015) investigate the impact of knowledge and stories shared across generations on innovation. Their study shows that in shared stories, a founder focus is negatively, and a family focus is positively, associated with innovation (Kammerlander *et al.*, 2015). When shared stories strongly focus on the founder, the family is typically more resistant to change in their business, resulting in less innovation (Kammerlander *et al.*, 2015). Family firms with a more salient focus on the family in their stories typically consider a broader range of potential options and are not necessarily constrained by past decisions (Kammerlander *et al.*, 2015).

5.2.1.2 Entrepreneurial orientation. KM processes in the family firm also affect the firm's EO. Drawing on stewardship theory and the knowledge-based view (KBV), for instance, Pittino *et al.* (2018) find that cross-generational knowledge sharing among family members is related to the dimensions of EO such that the effects on proactiveness and innovativeness are enhanced, yet the effect on risk-taking is negative. On one hand, the psychological

ownership of family members fosters knowledge sharing and the exchange of tacit knowledge, which improves the family firm's ability to identify and exploit entrepreneurial opportunities, enhancing the proactive and innovative posture of the business (Pittino *et al.*, 2018). On the other hand, the psychological ownership of kin also fosters the exchange of values and increases the family firm's desire to preserve their SEW, thus leading to a more conservative, risk-averse posture (Pittino *et al.*, 2018).

Further, the relationship between knowledge transfer and the family firm's EO is not always linear. Sciascia *et al.* (2013), for example, show that a curvilinear relationship exists between the number of family generations simultaneously involved in the top management team (TMT) and the family firm's EO. Drawing on upper echelon theory, Sciascia *et al.* (2013) note that the increased generational involvement in the TMT enhances knowledge transfer and the identification of entrepreneurial opportunities; however, as generational involvement in the TMT increases, family relationships become complicated, which impedes knowledge transfer and hinders the identification of entrepreneurial opportunities.

5.2.1.3 Internationalization. Internationalization is another outcome of family-firm KM that has received research attention. In a study by Fang *et al.* (2018), generational differences, in addition to available knowledge resources, are shown to affect the family firm's internationalization. These authors highlight that the founding generation often acts with future generations in mind and is motivated to create value for future generations by exploring alternative uses of knowledge-based resources. Given this, the founding generation is more likely to pursue strategic decisions, such as internationalization, as knowledge resources increase (Fang *et al.*, 2018). Moreover, Tsang (2020) theorizes that the family firm's internationalization process is constrained, in part, by the personal network of key family members given the limitations for broader knowledge transfer that supports the identification of foreign investment opportunities and also by the reluctance to hire outsiders to run overseas operations. These constraints, however, may be overcome by involving nonfamily members in the firm and engaging them in strategic decision-making.

Involving nonfamily members, though, is not a panacea given that the involvement of nonfamily members creates agency-related and an assortment of other issues (Chrisman *et al.*, 2014; Stewart and Hitt, 2012; Tabor *et al.*, 2018). Family firms, for example, can hire nonfamily members who have values that (at least partially) align with those of the family and/or hire individuals in (or based on recommendations from) their close social network to increase trust (Verbeke and Kano, 2012). In fact, in their study of international joint ventures, Debellis *et al.* (2021a) suggest that family firms (compared to nonfamily firms) have less strategic sensitivity – owing, in part, to the family firm's risk-averse posture in sharing knowledge in an international joint venture – that prevents them from engaging in such international alliances. A recommendation made for overcoming this strategic sensitivity is to open the board to external directors who meet the aforementioned criteria of having values that are (somewhat) aligned with the family and those with whom they have pre-existing social ties and trust (Debellis *et al.*, 2021a).

5.2.1.4 Financial performance and competitive advantage. As is common in firm-level studies, a number of investigations examine effects on firm performance and/or competitive advantage. Proper KM by the family firm, or any organization, has the potential to increase performance and competitive advantage. Studies reviewed generally confirm that the collective KM capabilities of the family firm positively influence firm performance in comparison to competitors (Hernández-Perlines *et al.*, 2017). More specifically, in a recent study by Santoro *et al.* (2021), the behavioral dimension of employee-level resilience is found to enhance the performance of small family firms because the owners are more likely to share their knowledge and vision, which builds commitment and resilience among employees and leads to enhanced knowledge application and, ultimately, increased competitiveness and performance.

Nonetheless, while the family firm has potentially valuable characteristics that enhance KM, firm performance, and competitive advantage, some characteristics hinder firm

performance and competitive advantage. For example, from a resource orchestration perspective, Chirico *et al.* (2011) note that when multiple generations are involved in a family firm seeking to be entrepreneurial, knowledge transfer is met with conflict, causing firm performance to suffer. To overcome such conflict, researchers find that when the family firm has multi-generational family involvement, coordinating mechanisms, like a participative strategy that increases involvement in knowledge transfer, can realize financial performance gains (Chirico *et al.*, 2011).

5.2.2 Family-centric outcomes. 5.2.2.1 Succession. Among the outcomes observed in this review, most studies examine firm-centric outcomes, yet some studies focus on family-centric outcomes. Among the family-centric outcomes studied, succession is one of the most commonly examined. The choice of the family firm's successor is partially attributed to the idiosyncratic nature of knowledge in the firm and is also influenced by the successful transfer of this knowledge across generations. For example, Lee *et al.* (2003) argue that family-firm knowledge is often specific and may be accessible to family members only; as such, the uniqueness of the family business and the transfer of such distinctive (tacit) knowledge to the next generation encourages the selection of a family successor over an outsider. Building on the work of Lee *et al.* (2003), research further shows the relative ease of transferring family-business-specific knowledge to an internal successor can make a family member a preferred successor over an external (nonfamily) successor (Royer *et al.*, 2008).

5.2.2.2 Transgenerational value creation and transgenerational entrepreneurship. Issues related to succession are not relevant if the family firm does not survive long enough to allow for the transfer of power. Thus, issues related to transgenerational value creation and transgenerational entrepreneurship represent other family-centered outcomes investigated in this literature. Family firms must create financial value over generations to survive long-term, and the ability to do so largely depends on the extent to which they can use and combine their current knowledge to generate new knowledge (Salvato and Melin, 2008). Such an achievement requires knowledge transfer across generations, and interestingly, in their study of transgenerational knowledge transfer and transgenerational entrepreneurship, Dou *et al.* (2020) find that elements of knowledge transferred across generations include moral values, competence values and cognitive heuristics, which highlight types of knowledge that are foundational to the transgenerational success of the family firm.

In addition, Chirico and Nordqvist (2010), in their study of dynamic capabilities and transgenerational value creation in family firms, suggest that knowledge and organizational culture can enable or inhibit the recombination of resources through which transgenerational value is created. Specifically, a closed and paternalistic family culture, in which knowledge transfer is not encouraged, fosters family inertia and negatively affects resource recombination and transgenerational value creation in family firms; whereas an open and entrepreneurial culture that encourages knowledge transfer and innovativeness counteracts family inertia and positively affects resource recombination and transgenerational value creation in family firms.

5.3 Moderators of knowledge management processes in family firms

Throughout the review, relationships between and among family-firm KM antecedents, processes and outcomes are reported. These relationships, however, vary given an array of contextual contingencies. Therefore, to offer insight into such contingencies, we note examples of key moderating effects that have been used to better understand the relationships between KM antecedents and processes, as well as the relationships between KM processes and outcomes.

5.3.1 Moderators of antecedent-knowledge management process relationships. 5.3.1.1 Family and nonfamily involvement. This review highlights a range of direct influences on the family firm's KM processes. However, the precise nature of these influences on KM processes are

typically complex and difficult to predict given the effects of contingency-based factors (Kotlar *et al.*, 2020). One category of moderating effects includes factors related to family and nonfamily involvement. Among family firms, the extent to which family are involved in firm-level governance varies and the level of involvement is likely to affect the firm's KM processes. One form of involvement shown to have a significant moderating effect is the family's involvement in the firm's management. Specifically, when the TMT is composed of a greater number of family members, fewer formal mechanisms are typically instituted, which enhances perceptions of empowerment and stewardship, leading to greater knowledge sharing when kin have psychological ownership (Pittino *et al.*, 2018).

Similarly, the involvement of nonfamily members varies across family firms. The result of their involvement depends, in part, on the relative number of nonfamily employees and the positions they hold. Nonetheless, involving nonfamily members can be instrumental to offsetting the insular nature of the dominant coalition of family, yet nonfamily involvement can have a hindering effect on the relationship between family social capital and knowledge application, as the family-related benefits that enhance knowledge transfer are mitigated when nonfamily members are involved (Daspit *et al.*, 2019).

5.3.1.2 Generational involvement. Furthermore, when kin are involved and embedded in the family firm, they are more likely to engage in stewardship behaviors that benefit the firm and, likewise, the KM processes. However, the extent to which this occurs is influenced by generational involvement. For example, Pittino *et al.* (2018) examine the psychological ownership of kin and find that while this type of ownership is positively related to knowledge transfer, the generation in control negatively moderates this effect. Specifically, the effect of psychological ownership on knowledge transfer is weaker in later-generation family firms while the same relationship is strengthened in earlier-generation family firms due to the closer, more informal relationships that facilitate knowledge transfer (Pittino *et al.*, 2018). Furthermore, based on insights from the KBV and family dynamics, Chirico and Salvato (2016) show that family firms run by later generations increase the positive effect of family social capital on knowledge application owing to the professionalization that manifests during later generations.

5.3.1.3 Environmental factors. Additionally, environmental factors affect the KM processes of the family firm. A recent study by Soluk *et al.* (2021) finds that the family's involvement in the firm is positively related to its knowledge application ability. Further, in this study, the dynamism of the environment is hypothesized to have a positive effect, thus strengthening the relationship between family influence and knowledge application given that when the environment is rapidly changing, the family-influenced firm is likely to dedicate "extra effort" toward knowledge application processes to survive uncertain times. Interestingly, though, the results suggest the opposite: when the environment is rapidly changing, the involvement of the family results in a weaker knowledge-application ability. Additionally, De Massis and Rondi (2020) note that the uncertainty introduced by the COVID-19 pandemic, which required businesses to quickly adapt to new constraints, is likely to limit the benefits of family and organizational social capital. Thus, external factors – especially those that require rapid adaptation – are likely to alter the effects of the positive and negative influences on family-firm KM processes.

5.3.2 Moderators of knowledge management process-outcome relationships. 5.3.2.1 Family and generational involvement in the firm. Prior investigations have also identified contingency factors that affect the relationship between KM processes and outcomes. Like the boundary conditions observed between antecedents and KM processes, some of the same factors are found to moderate the relationships with KM outcomes. For example, knowledge sharing practices enhance technological capabilities, yet the extent of the family's involvement in the firm and the generational involvement in management positively moderate the relationship between knowledge sharing and such technological capabilities (Zahra *et al.*, 2007). Involvement of multiple generations indicates a greater diversity of

knowledge shared, and the greater involvement of family members can support the development of family social capital, creating synergies for knowledge sharing that lead to enhanced technological capabilities.

5.3.2.2 Organizational social capital. Further, conceptualizing family social capital as a multi-dimensional construct, Herrero and Hughes (2019) find the effect of family social capital on the firm's financial performance depends on organizational social capital. Specifically, family firms with moderate levels of structural family social capital are argued to benefit from a high level of organizational social capital through the increased scope and likelihood of transferring knowledge from outside the family circle; thus greater organizational social capital enhances the positive effect of moderate structural family social capital on family firm performance (Herrero and Hughes, 2019). Overall, while the direct effects on KM and the effects of KM on outcomes are reported, such effects should be considered in the context of relevant boundary conditions.

6. Integrated insights and future opportunities

In this review, we offer insights into key antecedents, outcomes and moderators associated with family-firm KM processes based on a systematic review of 74 articles published in 23 journals. By conducting a comprehensive review of the literature, we identify key constructs and relationships examined to date, synthesize scholarly insights, and support advancement toward a less fragmented literature on family-firm KM processes. Continued scholarly progress, however, depends on the achievements of future researchers. Thus, based on this review, we offer a summary of insights gained and highlight promising opportunities for future investigation. As part of this effort, specific research questions are presented to elucidate particularly promising areas of future study with even further opportunities offered in Table 2.

6.1 Theory: insights and opportunities

The amount of research published on family-firm KM is increasing with impressive growth noted in the past decade [10]. This trend is not necessarily surprising, though, given that the field of family business research continues to experience rapid growth. As the number of studies on family-firm KM has increased, the theoretical frameworks [11] used have also evolved.

While some perspectives have continued to be used over time – like the KBV – recent studies have begun to incorporate more complex perspectives such as dynamic capabilities, absorptive capacity and behavioral perspectives. For example, a recent study by Soluk *et al.* (2021) integrates the dynamic capabilities perspective with family-firm-specific insights (e.g. strong attachment to the business, unique resources, risk aversion) to demonstrate that family influence is related to firm-level dynamic capabilities (e.g. knowledge application).

Additionally, among some of the most recent (and boundary-expanding) perspectives used is a family imprinting perspective to understand how the legacy of earlier generations shapes the approaches to innovation and tradition depending on what is imprinted on the current family generation (Erdogan *et al.*, 2020). Based on the approaches used by family firms to engage in innovation, Erdogan *et al.* (2020) find that when previous generations imprint family values and beliefs on the current generation, the current generation is more likely to adopt a preservation approach to their tradition, and the less previous generations imprint family values and beliefs on the current generation, the more the current generation is likely to adopt a revival approach to their tradition. This unique perspective advances our understanding of how family firms store/retrieve knowledge over time and shows how, in the family firm, knowledge from the past influences the present, potentially supporting "innovation through tradition." Given that many studies examine KM within a single generation of the family firm, studies that examine transgenerational knowledge transfer and

Table 2 Opportunities for future research on family-firm KM

Categories	*Future research directions*	*Examples of research questions*
Antecedents	Explore the effect of family social capital on various KM processes in family firms and further investigate the "dark side" of family social capital	How do the cognitive, structural and relational dimensions of social capital influence the development of family social capital, which thereby affects KM processes? What KM processes benefit the most from family social capital and which benefit the least? How can family social capital be best leveraged for its benefits while limiting the negative effects? What mechanisms can be strategically used to offset the negative effects of social capital? How does a micro foundational perspective help to understand the development and subsequent effect of family social capital on family members, the family firm and other stakeholders?
	Explore the effect of SEW dimensions on various KM processes in family firms	How can the family's desire to maintain control over the firm be preserved for its beneficial effects while minimizing the risk of losing control when (knowledge) resources are needed from nonfamily actors? How, when and to what extent do conflicting family and business goals affect family-firm KM processes? Which noneconomic goals enhance, and which restrict, the development of KM processes in the family firm? Do restricted (versus extended) SEW goals always limit the development of KM processes? In what conditions might the family's short-term, family-centered, noneconomic goals enhance the KM processes of the family firm?
	Explore the role of family business culture on KM processes in family firms	What type of family-firm culture is best-suited for developing, leveraging and sustaining KM processes that involve family and/or nonfamily stakeholders both internal and external to the firm? What is the best approach for integrating nonfamily employees into a family-firm culture focused on KM and innovation? Does the culture that exists within the controlling family "spillover" into the family firm and affect KM processes? If so, how and when?
	Explore approaches family firms use to store and preserve their identity and memory	What strategic, long-term approaches do family firms use to preserve their heritage while retaining their organizational and family memory? What is the process through which the long-surviving family firm is able to effectively recall and reinterpret stored knowledge to successfully innovate through tradition? What effect do emotions, memories, ambitions, and other psychological factors have on the stories developed and transmitted about the family firm?
	Explore the influence of family-governance structures on KM processes	Is there an optimal governance configuration to enhance family-firm KM processes? If so, to what extent do conditional factors, such as firm generation and nonfamily involvement, have an effect? As the family firm professionalizes and involves nonfamily employees, how can bifurcation bias be mitigated to avoid factions that inhibit knowledge transfer and other KM processes? Understanding that the financial and nonfinancial goals of family firms different in type and the extent pursued, for those outcomes related to the KM process, what governance configurations are ideal for the differing configurations and variation among such goals?
	Take a micro foundational perspective to understand KM processes in family firms	What founder personalities impede knowledge transfer to the incumbent? What insights from psychology can be used to enhance the integration of nonfamily members into the family firm for the purpose of facilitating more efficient knowledge creation, storage/retrieval, transfer and application? What mechanisms can be used to overcome knowledge-transfer hurdles when family-based relationships are not based on open, generalized exchanges?
	Explore how KM processes affect one another and the effects of family influence on KM processes	How are the KM processes related within the family firm? Do the relationships among the processes differ when compared to nonfamily firms? If so, how? What are the relationships among KM processes and knowledge-related dynamic capabilities? How do KM processes affect lower-order capabilities and how do higher-order capabilities affect KM processes?
	Explore the effect of crises on KM processes in family firms	To what extent and how do crises (e.g. the COVID-19 pandemic) affect the KM process in family firms? What can family firms do to safeguard their KM processes when faced with unexpected, turbulent events?

(*continued*)

Table 2

Categories	*Future research directions*	*Examples of research questions*
Outcomes	Take a multi-level approach to examine the effect of KM processes on outcomes at another level	How does the knowledge transfer within the firm affect individual-level perceptions and behaviors requisite for future knowledge creation, transfer, etc.? What, if any, community-level outcomes (especially communities with greater concentrations of family firms) are affected when family firms excel at or are limited in, their KM abilities?
	Examine the effect of knowledge storage on outcomes important to the family firm	How does the preservation of knowledge and tradition relate to outcomes such as strategic flexibility, internationalization and family wealth? In what contexts is "innovating through tradition" the most beneficial for achieving (non) financial goals?
	Examine the effect of transgenerational knowledge transfer on outcomes important to the family firm	What individual, group/family and firm-level characteristics best support the transgenerational exchange of knowledge? In what situations may the transfer of knowledge across generations not be beneficial for family and/or firm-specific outcomes?
	Examine the impact of nonfamily-employee knowledge on family-firm outcomes	How and when does knowledge transferred from nonfamily members affect family firm succession? What mechanisms can facilitate/hinder this transfer? What are the most effective means for transferring knowledge between family and nonfamily members given the proliferation of virtual communication options, and what effect does the use of virtual-communication technology have on family-firm outcomes?
	Examine mediation-based and longitudinal effects	From a process model perspective, what is the sequential relationship between salient influences, KM processes and related family-firm outcomes? In addition to the antecedents, KM processes and outcomes observed in this review, what other factors potentially mediate the relationships involving KM processes? Under what conditions do these relationships vary?
Moderators	Examine boundary conditions at multiple levels that affect KM processes	What individual-level factors (e.g. founder personality) moderate the relationships between/among antecedents, KM processes and/or outcomes? Why and when? What family-level factors (e.g. family units plagued with instability and/or a lack of cohesiveness) moderate the relationships between/among antecedents, KM processes and/or outcomes? Why and when? What industry-level factors (e.g. industry munificence) moderate the relationships between/among antecedents, KM processes and/or outcomes? Why and when? What institutional-level factors (e.g. informal regulations) moderate the relationships between/among antecedents, KM processes and/or outcomes? Why and when?

how, across generations, knowledge of traditions can be stored and recalled for innovative purposes are well-positioned to offer valuable contributions:

RQ1. How is knowledge transferred, stored and recalled across generations, and how can the preservation of tradition-based knowledge be most effectively used to deliver innovative outcomes for family firms across industries, time and contexts?

Collectively, the studies examined note that certain factors yield a distinct advantage, and some simultaneously impose a liability, to family-firm KM. For instance, the commonly examined resource of family social capital is noted to exert conflicting effects on certain KM processes. Whether family social capital yields a positive or negative effect depends, in part, on the family members engaged and the nature of their relationships. Although family social capital and related family dynamics are commonly researched in family firms, we are yet to fully understand what, when and how individual-level factors drive group and firm-level behaviors.

Following the call of De Massis and Foss (2018), the use of a microfoundational perspective is likely to yield valuable insights by fostering investigation into the interrelated effects of the emotions, memories and experiences of family (and nonfamily) members in the family firm. Relational bonds – even those steeped in trust and developed over time – are not impenetrable. As noted by Humphrey *et al*. (2021), exploring the psychological foundations of (knowledge) management in family firms will offer clarity into how and why strategic

behaviors differ from nonfamily firms and highlight differences among family firms. To support this pursuit, the integration of theories and insights from psychology are needed to better understand the relationships among emotions, memories and experiences and to gain insight into how these elements affect collective dynamics that result in firm-level processes such as KM:

RQ2. What theories and insights from psychology (and related disciplines) can be applied to advance our understanding of family-firm KM antecedents, processes, outcomes and moderators, and how do individual-level cognition, emotions and experiences affect group/family dynamics and family-firm behaviors?

6.2 Antecedents: insights and opportunities

An overview of the antecedents studied highlights the importance of trust-based relationships to family firms and their KM processes. For each KM process, nearly all antecedents either directly (or indirectly) relate to trust. For example, family social capital is founded in a deep level of trust that develops over a long period of time among family members. When this level of trust exists, the family firm is able to enhance its knowledge creation, knowledge transfer and knowledge application. However, there is a downside: the trust that resides within family social capital also creates an insularity among kin, limiting the firm's knowledge creation and knowledge transfer. Family social capital is shown to be imperfectly imitable by nonfamily members (Herrero, 2018) and exists in at least three broad configurations (Sanchez-Ruiz *et al.*, 2019). While these and other studies advance our understanding of family social capital, how can family firms achieve optimal value creation wherein the benefits of family social capital are maximized and the negative effects minimized?

RQ3. How can family social capital be optimally leveraged by the family firm to deliver maximum value creation for each type of knowledge process?

Each antecedent also relates, in some form, to the family. The family is a source of difference among family firms. The dynamics of the family are regularly noted as influential to KM processes. For example, Zahra (2012) notes the importance of family cohesion, with more cohesive families having enhanced capabilities with respect to some KM processes, offering evidence that internal family dynamics matter. Given this importance, a refined understanding of the family will allow for a more robust understanding of the family firm. Jaskiewicz and Dyer (2017) note the following four common dimensions of families that are derived from family science theories: family structures, family functions, family interactions and family events. Along this line, to better understand KM processes and related differences among family firms, more remains to be known about the differences among families. Jaskiewicz and Dyer (2017) note numerous opportunities for integrating family science theories [12] into management, and accordingly, we look forward to a better understanding of the differences among families given that such insights will have valuable implications for understanding what, how, when and why family firms strategically develop and leverage KM processes:

RQ4. How do differences in and among families, as well as varied definitions of "family," affect family-firm KM processes?

Family firms tend to pursue a broader set of outcomes than nonfamily firms (Holt *et al.*, 2017). Developing and maintaining SEW is a salient noneconomic goal for many (perhaps all) family firms (Berrone *et al.*, 2012). However, the scope of noneconomic goals is expansive for families and their firms, and investigating the effects of these goals on KM processes presents numerous opportunities. Particularly, future researchers can consider the effects of the family's desire to maintain control over the firm and the extent to which (and how) this desire affects KM processes. Maintaining control allows the family firm to prioritize desired goals; however, questions related to which noneconomic goals enhance

(and which restrict) the development of KM processes remain to be answered. Following the work of Miller and Le Breton-Miller (2014), extended SEW goals may support the superior growth and development of KM processes, yet do restricted SEW goals, which are focused on more immediate outcomes for the family, always hinder KM processes? If not, in what conditions might the family's short-term, family-centered, noneconomic goals enhance the KM processes of the family firm? Studies suggest that conflicts may arise among family and business priorities, giving rise to a potential "dark side" of family influence (Biscotti *et al.*, 2018). Although studies have begun to examine the effects of family-centered, noneconomic goals, more work is needed to determine what, how and when KM processes enable a family firm to achieve these desired outcomes. Moreover, expanding the scope of outcomes examined – to include a broader range of nonfinancial outcomes – will offer greater clarity on the types of outcomes pursued and the extent to which they are achieved:

RQ5. What noneconomic goals affect KM processes, how do KM processes allow family firms to achieve noneconomic outcomes, and how does the potential conflict between financial and nonfinancial pursuits affect investments in family-firm KM?

6.3 Outcomes: insights and opportunities

Reviewing family-firm KM studies also presents insights and opportunities for future research related to the outcomes of KM processes. Prior research highlights the importance of protecting and preserving the family's identity (Magistretti *et al.*, 2020). Some family firms use branding strategies to communicate the family and corporate heritage via their company website as a strategy to preserve their family identity (Micelotta and Raynard, 2011). Future research is needed to examine what other approaches are successfully used by family firms (and families) to preserve their heritage and imprint an entrepreneurial legacy (Jaskiewicz *et al.*, 2015).

Family firms may seek to preserve the beliefs and knowledge that constitute their traditions, yet, at the same time, family firms must innovate to remain competitive. Extending the work of De Massis *et al.* (2016), which examines family firms that innovate through tradition, Erdogan *et al.* (2020) note that innovation and tradition create a paradox. They note that family firms can successfully navigate the paradox through a specific capability. Building on these insights, how can family firms remain competitive while preserving their organizational memory and traditions? It is noted that the storage and recall of knowledge is essential, yet much remains to be understood about the processes through which longer-surviving family firms are able to effectively store, recall, reinterpret and apply knowledge to innovate through tradition. Understanding these knowledge processes holds noteworthy promise:

RQ6. What is the process through which family firms are able to engage in activities like "innovating through tradition" to enhance transgenerational survival and success?

In this review, some antecedents are observed to offer both positive and negative effects on KM processes (e.g. family social capital, SEW preservation intentions). Additionally, KM processes are noted to have varied effects on outcomes. However, more comprehensive studies are needed to examine the longitudinal and multi-level effects that antecedents have on KM processes, which thereby yield categorically different and varied levels of outcomes.

Furthermore, work that examines multiple KM processes simultaneously is particularly interesting. A recent study by Muñoz-Bullón *et al.* (2020) empirically demonstrates that family firms using a combined innovation strategy – wherein innovation activities occur both in the firm and are outsourced to third parties – achieve greater innovation outcomes compared to nonfamily firms with similar strategies. The family firm's ability to gain this type of competitive advantage is attributed to being selective and establishing trust-based relationships with external partners (Duran *et al.*, 2016), their organizational and family social capital (Arregle *et al.*, 2007; Sanchez-Ruiz *et al.*, 2019), noneconomic goals such as

reputation (Berrone *et al.*, 2012) and related factors. Insights like these, which take a more holistic approach to understanding the (paradoxical) effects on KM processes, the simultaneous use of multiple KM processes, and the outcomes obtained as a result are likely to offer notable advances to family-firm KM scholarship:

RQ7. What and how do multi-level factors affect family-firm KM processes, how are paradoxical challenges related to KM overcome by family firms and used to create a competitive advantage, and what (non)economic outcomes are generated over time?

6.4 Moderators: insights and opportunities

The relationships among antecedents, KM processes and outcomes vary given relevant boundary conditions. Our review identifies moderating effects that are commonly examined across studies, and among these, the extent of the family's involvement is commonly observed. Even though the involvement of the family leads to potentially valuable and unique resources and capabilities, the family firm composed only of kin may, at some point, need to hire nonfamily members to grow and/or acquire the resources needed to survive (Stewart and Hitt, 2012). In family firms with heightened family social capital, the involvement of nonfamily members is noted to enhance the ability of the firm to acquire and transfer external knowledge; however, nonfamily members also mitigate the benefits of family social capital for knowledge application purposes. Given that hiring nonfamily members is likely to be required for many family firms, and because the vast majority of family firms employ nonfamily members (Tabor *et al.*, 2018), how can jobs for nonfamily (and family) be designed so that maximum value is created for and from the family firm's KM processes?

RQ8. Leveraging insights from human resource management, what is the optimal job design for the relevant positions in the family firm that yields the greatest benefit to KM processes and provides the maximum attainment of desired outcomes?

6.5 Additional insights and opportunities

Knowledge is typically created through collaborative mechanisms (Nonaka, 1994). In comparison to nonfamily firms, family firms often provide family members with opportunities for social interactions. Of the studies reviewed, several compare family and nonfamily firms with respect to the antecedents and outcomes of their KM processes. For instance, Debicki *et al.* (2020) investigate the strategic approaches of family and nonfamily firms and the effects on relational capital, finding that unique family resources enhance the firm's knowledge application.

There is a substantially greater contingent of studies, however, that examines family firm heterogeneity [13]. The differences among family firms have been examined by exploring different levels of family ownership (Kotlar *et al.*, 2020; Sciascia and Mazzola, 2008; Zahra, 2012), family involvement in the TMT (Pittino *et al.*, 2018), generational involvement (Chirico *et al.*, 2011) and nonfamily member involvement (Randolph *et al.*, 2021), etc. Although the majority of studies examined investigate differences among family firms, much remains to be understood about family-firm heterogeneity. Daspit *et al.* (2021: 3) define family-firm heterogeneity as "the range of categorical and/or variational difference(s) between or among family firms at a given time or across time." Based on this definition, what are the various facets of difference that exist among family firms with respect to KM processes? Additionally, how do these differences evolve over time?

RQ9. What are the categorical, variational, spatial and temporal differences among family firms with respect to their KM antecedents, processes and outcomes?

Finally, studies related to KM in general (not specific to family firms) have appeared in an array of journals, crossing disciplinary boundaries. Given this scope, numerous theoretical perspectives have been used to examine KM. Specifically, Matayong and Mahmood (2013) reviewed KM literature and identified these nine theories that are the most common theoretical frameworks of general KM studies: innovation diffusion theory, the theory of reasoned action, the technology acceptance model, the theory of planned behavior, task-technology fit, social capital theory, social cognitive theory, the IS success model and the IS continuance model. For the study of KM to move forward, Paoloni *et al.* (2020) note that KM research will benefit from recognizing and more closely examining the role of social capital and related factors.

Looking forward, KM researchers may benefit by integrating insights from family-firm scholarship given the unique relational dynamics that manifest in this context. With the coupling of the family and business systems, unique (social) resources and capabilities are created (Habbershon and Williams, 1999; Habbershon *et al.*, 2003). In their seminal work, Pearson *et al.* (2008) take a social capital perspective of familiness and elucidate the unique structural, cognitive and relational resources, as well as the unique capabilities (i.e. refined knowledge and associability capabilities) that create idiosyncratic value for the family firm. However, while the social capital created among family is "imperfectly imitable" by nonfamily firms (Herrero, 2018), not all family firms have similar levels of family social capital (Sanchez-Ruiz *et al.*, 2019). Therefore, given that family firms are the most common form of business, KM researchers are encouraged to consider the extent of family involvement in the organizations studied and account for family-specific influences, like family social capital, to obtain more precise insights.

Further, Durst and Edvardsson (2012) note in their review of KM that more work is needed to address both the sender and the receiver involved in the knowledge-transfer process (many studies focus on one side). Here too, the study of family firms can be potentially helpful given that family-firm scholars use an array of perspectives to understand the involvement of nonfamily members in the family firm. For instance, agency, stewardship, social capital, expectancy, commitment, psychological ownership and other perspectives have been employed to examine nonfamily members (see review by Tabor *et al.*, 2018). These, and other, perspectives can be used to better understand both sides of the knowledge-transfer process and more accurately account for differences between the actors engaged in the exchange:

RQ10. What perspectives and insights can be leveraged from family-firm scholarship to advance the study of KM more broadly, and what perspectives from KM can be leveraged to advance family-firm scholarship?

6.6 Limitations

Although this review contributes to both family-business and KM scholarship, the contributions are not without limitations. First, we examine family-firm KM literature by categorizing the articles based on four KM processes including knowledge creation, storage/retrieval, transfer and application. While this conceptualization is consistent with the seminal work of Alavi and Leidner (2001), we recognize that other conceptualizations of KM processes exist and are relevant. Nonetheless, by basing the review on a well-accepted conceptualization, we organize insights in a manner that can be broadly understood. Second, in the review, we identify key antecedents, outcomes and moderators; however, we elected to not review the relationships between and among KM processes. Although some studies examine such relationships, we forego such detail for the purpose of parsimony. Nonetheless, when an identified study examined more than one KM process, each process was coded, and when possible, we focused the review on the primary aspect of KM examined. Third, while we review an array of factors, the intention is not to reference every construct and relationship identified in the literature. Instead, we note key factors and

represent the general nature of the relationship(s) observed while fully acknowledging that differences in context and magnitude create differing effects. Fourth, given the expansive nature of the review, we purposefully reserve methodological observations. Method-related advances are, indeed, necessary to the advancement of knowledge on family-firm KM processes, and we look forward to work that offers promising ways to enhance the empirical study of family-firm KM processes.

7. Conclusion

In this review, we aim to reduce the fragmentation in the family-firm KM literature. Based on insights from 74 articles published in 23 journals, we coalesce findings into a framework that highlights antecedents, outcomes and moderators related to KM processes in the family firm. Additionally, we offer a summary of integrated insights and identify promising opportunities for future research with the intention of helping to advance scholarly work in the fields of family business and KM, both respectively and in a complementary manner. While much work remains to be done, we look forward to future insights that will advance our understanding of KM in family firms.

Notes

1. Following Alavi and Leidner's (2001) seminal review article on KM, we adopt a process view of KM.
2. A prominent scholarly focus on understanding family-firm innovation is, of course, only one example of why less research attention has focused on family-firm KM.
3. The following keywords were used: *knowledge manag**, *km*, *knowledge shar**, *knowledge transfe**, *knowledge proces**, *knowledge creat**, *knowledge innovat**, *knowledge accumulat**, *knowledge stor**, *knowledge system*, *knowledge acqui**, *knowledge assimilati**, *knowledge exploit**, *knowledge use*, *knowledge explor**, *knowledge applicat**, *absorptive capacity*, and *intellectual capital*, coupled with *family business**, *family own**, *family control**, *family influence**, *family involve**, *family firm**, *firm enterprise** and *family manage**.
4. Scopus is claimed to be the largest abstract and citation database (Kellens *et al.*, 2013), and WoS is known for its objective journal-selection criteria and prolific use in the academic community (Perri and Peruffo, 2016).
5. We appreciate one of the anonymous reviewers for this suggestion.
6. We followed the same search technique (that was not date restricted) by using the same keywords in Scopus as in the "topic" of WoS. In WoS, the "topic" includes a search of the article title, abstract, author keywords and KeyWords Plus®. KeyWords Plus® terms are index terms automatically generated from the titles of cited articles and include terms that appear more than once in the references of a given article. Given that WoS includes KeyWords Plus® in search output, this is the most likely explanation for why the number of articles identified from the WoS search is greater than the number identified from the Scopus search. (More information on KeyWords Plus® can be found at https://images.webofknowledge.com/images/help/WOS/hp_full_record.html.)
7. At least one of the following KM processes was required to be examined in the study for the article to be included in the review: knowledge creation, knowledge storage/retrieval, knowledge transfer or knowledge application. KM processes that used other names for these processes were included when appropriate.
8. Articles reviewed are published in the following journals, and the number of articles identified in each journal is noted in parentheses; however, journals from which no relevant articles were found are not reported: *Academy of Management Review* (1), *Asia Pacific Journal of Management* (2), *British Journal of Management* (1), *Business History* (2), *Entrepreneurship and Regional Development* (1), *Entrepreneurship Theory and Practice* (7), *Family Business Review* (15), *Global Strategy Journal* (1), *International Business Review* (1), *International Small Business Journal* (5), *Journal of Business Research* (6), *Journal of Business Venturing* (1), *Journal of Family Business Strategy* (9), *Journal of Financial and Quantitative Analysis* (1), *Journal of International Management* (1), *Journal of Knowledge Management* (6), *Journal of Product Innovation management* (3), *Journal of Small Business Management* (2), *Journal of World Business* (1), *Psychology and Marketing* (1), *Small Business Economics* (5), *Strategic Entrepreneurship Journal* (1) and *Strategic Management Journal* (1).

9. Terms are italicized to indicate a common construct observed across studies reviewed. Italicized constructs, such as "family social capital," are featured in the integrated conceptual map offered in Figure 2.
10. In total, 57 out of 74 articles were published between 2011 and 2021, representing 77% of the articles reviewed.
11. The most common conceptual frameworks observed in this review of family-firm KM scholarship include the KBV (Bouncken *et al.*, 2020), the RBV (Bamel and Bamel, 2018; Zybura *et al.*, 2021), absorptive capacity (Brinkerink, 2018; Kotlar *et al.*, 2020) and the dynamic capability perspective (Barros *et al.*, 2016). These perspectives, which are central to many studies, are typically integrated with perspectives commonly found in family business research (e.g. SEW, familiness, agency theory) to understand KM processes in family firms.
12. In their review, Jaskiewicz *et al.* (2017) highlight seven prominent family science theories, note their main constructs and predictions, and provide insights for how each can be integrated into management scholarship. The family science theories identified include (in the order presented in their work): family systems theory, family communication patterns theory, family-niche model of birth order and personality, parental control theory, family development theory, ABCX model of family stress and intergenerational solidarity theory.
13. In total, 57 out of 74 articles examine heterogeneity among various types of family firms, representing 77% of the total number of articles reviewed.

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